Exit Full Screen View

Problem

Submissions

Leaderboard

Discussions

Given a matrix of size N × M, print all elements of the matrix in spiral order starting from the top-left corner.

The traversal order is:

Left to Right across the top row Top to Bottom along the right column Right to Left across the bottom row Bottom to Top along the left column

Continue this process until all elements are visited

Input Format

The first line contains two integers N and M, representing the number of rows and columns. The next N lines contain M space-separated integers representing the matrix.

Constraints

1≤N,M≤1000 −10 9 ≤Matrix[i][j]≤10 9

Output Format

Print the elements of the matrix in spiral order separated by spaces.

Sample Input 0

```
3 3
1 2 3
4 5 6
7 8 9
```

Sample Output 0

```
1 2 3 6 9 8 7 4 5
```

Sample Input 1

```
4 4
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
```

Sample Output 1

```
1 2 3 4 8 12 16 15 14 13 9 5 6 7 11 10
```

Change Theme Language Python 3

```
1  n, m = map(int, input().split())
2
3  a = []
4
5  for i in range(n):
6      a.append(list(map(int, input().split())))
7
8  top = 0
9  bottom = n - 1
10 left = 0
11 right = m - 1
12
13 ans = []
14
15 while top <= bottom and left <= right:
16
17     for j in range(left, right + 1):
18         ans.append(a[top][j])
19     top += 1
20
21     for i in range(top, bottom + 1):
22         ans.append(a[i][right])
23     right -= 1
24
25     if top <= bottom:
```

Line: 36 Col: 1

Upload Code as File

Run Code Submit Code

Test against custom input

Test case 0

Test case 1

Compiler Message

Success